SYED MAOZ ARIF

Hardware & Embedded Systems Engineer

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RESEARCH PROFILE

Electrical and Computer Engineering student specializing in embedded systems, custom silicon (FPGA) architecture, and edge computing. Proficient in hardware description languages (Verilog), mixed-signal PCB design, and IoT-driven digital twins. Recognized globally for precision engineering (Instructables Sensors Contest) and awarded multiple national positions for solving complex engineering tasks. Passionate about contributing to cutting-edge scientific research, hardware optimization, and advanced electronics at the Max Planck Society.

EDUCATION

B.S. Electrical Engineering — Expected 2028

COMSATS University Islamabad, Abbottabad Campus — GPA: 3.61 / 4.0 (5th semester)

- Focus Areas: Embedded Systems, Digital Logic Design, PCB Architecture, Signal Integrity
- Altium Global Scholarship Awardee (March 2026)

KEY ENGINEERING & RESEARCH PROJECTS

Stratum-1 Precision Hardware Clock & GPS Bridge — August 2026

Verilog | Lattice iCE40UP5K FPGA (iCEBreaker) | Digital Logic | NEO-6M GPS | NMEA

- Engineered a Stratum-1 precision hardware clock synchronizer using pure digital logic, completely bypassing the latency of microprocessors and software operating systems.
- Developed a custom hardware UART receiver in Verilog to sample 9600 baud serial data directly from the GPS module.
- Engineered a robust state machine to parse the live NMEA stream, isolating atomic time sentences to pulse a hardware LED in absolute synchronization with satellite atomic clocks.
- Overcame complex Physical Constraint File (PCF) routing challenges by dynamically remapping internal oscillator clocks.

Industrial Edge-AI Condition Monitoring & 3D Digital Twin — July 2026

STM32H7 | ESP32 WROOM | Panasonic Relays | Altium Designer | Ethernet

- Designed a 1st Position-winning industrial-grade development board featuring dual-MCU architecture, hardware calibration, and 70 GPIOs for advanced edge AI and Digital Twin applications on 3-Phase Induction Motors.
- Achieved robust 24V power handling and real-time telemetry extraction for complex diagnostic AI models.

ElectroParts IMS: Offline-First Embedded Component Architecture — August 2026

JavaScript (ES6+) | Firebase Realtime DB | CapacitorJS | JsBarcode

- Architected a cross-platform (Web/Native Android) inventory engine supporting 32 bespoke electronics categories to manage thousands of integrated circuits, microcontrollers, and precision passives.
- Implemented dual-storage local/cloud sync and integrated linear barcode generation algorithms for zero-latency retrieval.

IncliSense Precision Digital Level — April 2026 | **Global Runner-Up, Instructables**

Arduino Nano | MPU6050 | 3D Printing | I2C | Spatial Telemetry

- Developed a precision leveling device analyzing dual-axis spatial data via I2C, yielding high-accuracy numerical feedback with PC telemetry processing. Published detailed engineering documentation for the global maker community.

High-Frequency & Power Management Systems (Mixed-Signal PCBs) — 2025 – 2026

Altium Designer | Impedance Control | MOSFETs | BQ76930

- **RF Transceiver Board:** Designed a mixed-signal ESP32 PCB with 50Ω impedance-matched RF traces and a unified ground plane utilizing 312 stitching vias for a Faraday cage effect.
- **6S UAV BMS:** Developed a 6-cell battery management system schematic ensuring high-current safety via specialized charge/discharge MOSFET configurations and hardware protection.

AWARDS & HONORS

- **2nd Position – EMCOT 2026 Best Complex Engineering Project** - COMSATS University & IET August 2026
- **1st Position – Best CEP Project Award (Software Category)** - COMSATS University & District Youth July 2026
- **Runner-Up – Instructables Sensors Contest 2026 (Global)** - Bronze Medal for 'IncliSense" project April 2026
- **Altium Global Scholarship Awardee** - Selected for emerging engineering excellence March 2026

- **1st Position – Best Circuit Design -** COMSATS University & IET Hardware Workshop Feb 2026

PROFESSIONAL EXPERIENCE & LEADERSHIP

President (Previously Vice President & Membership Head) May 2025 – Present
IET On Campus, CUI Abbottabad Leadership

- Directing student branch operations, promoting academic research excellence, and spearheading technical initiatives.
- Organized specialized workshops including 'Computer Vision & AI in UAVs" and PCB fabrication seminars.

Freelance Hardware & Mechanical Designer Ongoing
Fiverr & Upwork Professional

- Conducting professional multi-layer PCB design, schematic capture, and 3D CAD modeling for international clientele.

Cabinet Member 2025 – 2027 Session
Youth Parliament Pakistan Governance

- Selected for national advocacy and policy analysis representing the youth engineering demographic.

TECHNICAL SKILLS

Hardware Design	Altium Designer, CircuitStudio, KiCad, Proteus, LTspice, Multisim
Embedded Systems	FPGA (Lattice), ESP32, STM32, Arduino, ATMEGA, I2C, SPI, UART, RF Control
Programming	Verilog, C/C++ (Embedded), Python, JavaScript (ES6+), HTML5, CSS3, MATLAB
Software/Cloud	CapacitorJS, Firebase, Firebase Auth, Git, iceStorm Toolchain, LaTeX
3D CAD & Modeling	Autodesk Inventor, SolidWorks, Fusion 360, AutoCAD, 3D Printing